Aggregate Demand and Supply Analysis

Chapter 22

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview and Learning Objectives

This chapter introduces the AD–AS framework to analyze how monetary policy affects output and prices.

Key question: why do output and inflation fluctuate over the business cycle, and how does the economy return to potential output?

Outline

Business Cycles and Inflation

Aggregate Demand

Aggregate Supply

Short-Run and Long-Run Equilibrium

Aggregate Demand Shocks

Supply Shocks

Conclusions from AD–AS Analysis

Business Cycles and Inflation

Business Cycles are fluctuations in overall economic activity, usually measured by changes in **real GDP**.

Main Phases

- **Expansion**: GDP rises, unemployment falls, income increases.
- **Peak**: Highest level of economic activity.
- **Recession**: GDP declines and unemployment rises.
- **Trough**: Lowest point before recovery begins.

Example

The **2007–2009 Global Financial Crisis** led to a severe global recession.

Measuring Business Cycles

Indicators

- **Real GDP**: Total output adjusted for inflation.
- **Unemployment Rate**: Share of workers actively seeking jobs.
- **Output Gap**: Difference between actual and potential GDP.

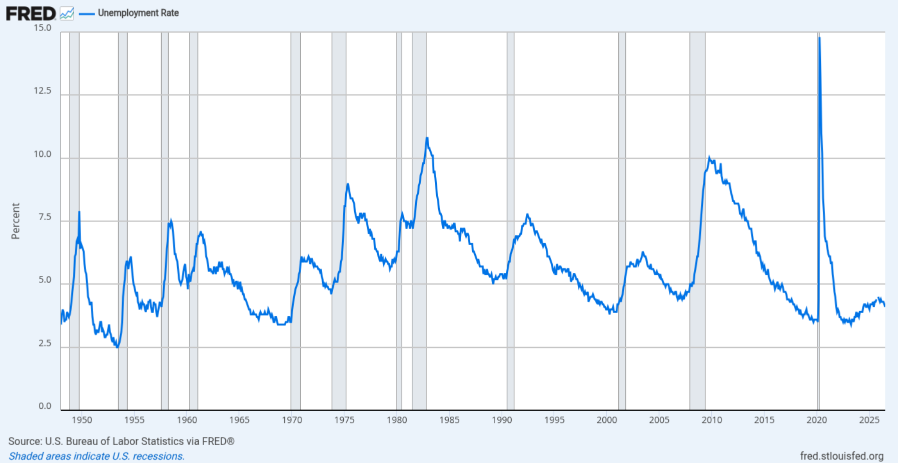
Output Gap

- **Negative Gap**: Economy below potential (recession).
- **Positive Gap**: Economy above potential (overheating).

Example

In **2020 (COVID-19)**, the U.S. output gap fell to about **-10%**.

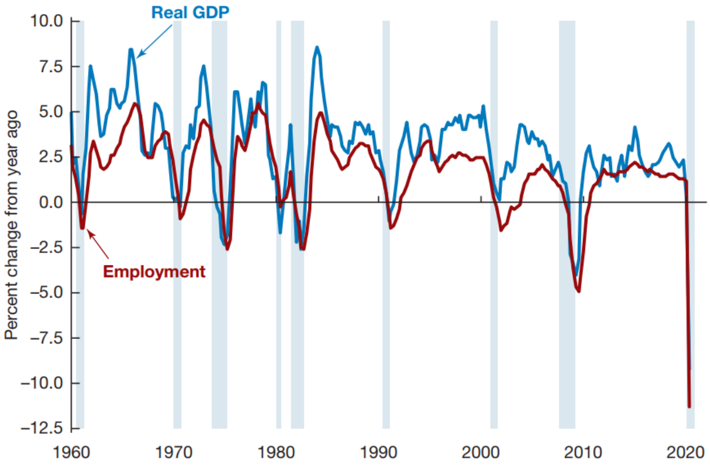
The U.S. Unemployment Rate Since 1948



Unemployment rises in every recession (shaded) and falls during expansions; note the 1982 and 2009 peaks and the 2020 spike.

Source: FRED, Federal Reserve Bank of St. Louis

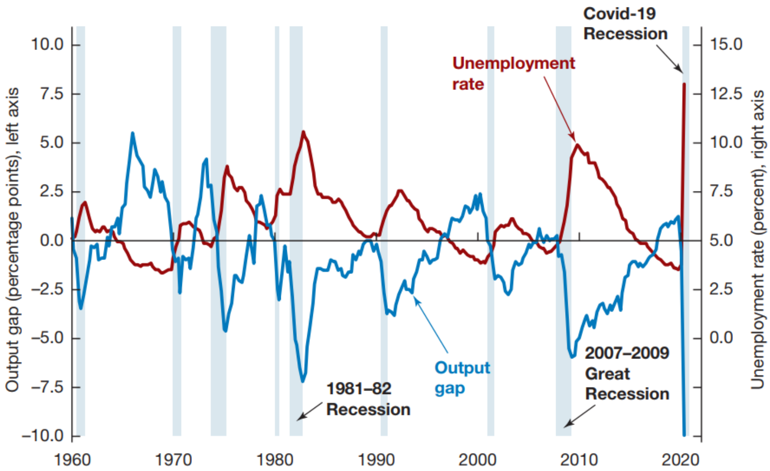
Real GDP and Employment over the Business Cycle



Growth rates of U.S. real GDP and employment, 1960–2020; recessions shaded.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Output Gap and Unemployment over the Business Cycle



U.S. output gap and unemployment rate, 1960–2020: a negative gap goes with high unemployment.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

What is Inflation?

Inflation is the rate at which the **general price level** of goods and services rises.

- **Consumer Price Index (CPI)**
- **GDP Deflator**

Types of Inflation

- **Demand-Pull**: High demand pushes prices up.
- **Cost-Push**: Higher production costs raise prices.
- **Hyperinflation**: Extremely rapid price increases.

Example

U.S. inflation exceeded **14% in 1980** due to oil shocks and rapid money growth.

Inflation over the Business Cycle

Inflation often moves with the business cycle.

Across the Cycle

- **Expansion**: Rising demand increases inflation.
- **Recession**: Falling demand lowers inflation or causes deflation.

Great Recession (2007–2009)

- Inflation fell sharply; some periods experienced deflation.
- The Fed introduced **Quantitative Easing (QE)**.

Inflation and Unemployment

The relationship between inflation and unemployment is described by the **Phillips Curve**.

Phillips Curve

- **Short Run**: Lower unemployment $\rightarrow$ higher inflation.
- **Long Run**: The trade-off weakens as expectations adjust.

Examples

- **1970s**: Stagflation (high inflation + high unemployment).
- **1990s–2000s**: Low inflation and low unemployment.

Policymakers must balance inflation control and employment.

Policy Responses

Monetary Policy

- Central banks adjust interest rates.
- **Expansionary**: Lower rates stimulate demand.
- **Contractionary**: Higher rates reduce inflation.

Fiscal Policy

- Government spending and taxation influence economic activity.
- **Stimulus**: Increase demand (e.g., COVID-19 relief).
- **Tax hikes / spending cuts**: Reduce inflation pressure.

Paul Volcker, Fed chair 1979–1987, whose rate hikes ended the high inflation of the 1970s at the cost of a recession.

Example

Outline

Business Cycles and Inflation

Aggregate Demand

Aggregate Supply

Short-Run and Long-Run Equilibrium

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Supply Shocks

Conclusions from AD–AS Analysis

Aggregate Demand

Components

- **Consumption (C)**: Household spending on goods and services
- **Investment (I)**: Business spending on capital and new housing
- **Government Purchases (G)**: Government spending on goods and services
- **Net Exports (NX)**: Exports minus imports

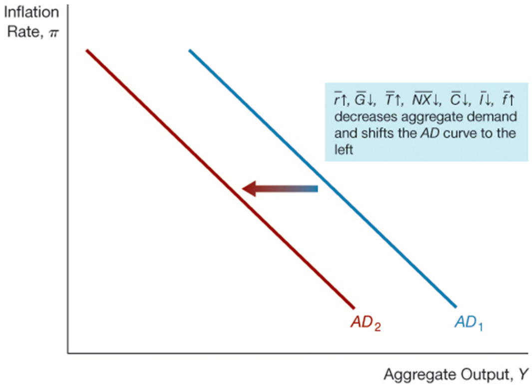
$$Y^{ad} = C + I + G + NX$$

Why the AD Curve Slopes Downward

$$P \downarrow \Rightarrow \frac{M}{P} \uparrow \Rightarrow i \downarrow \Rightarrow I \uparrow \Rightarrow Y^{ad} \uparrow$$

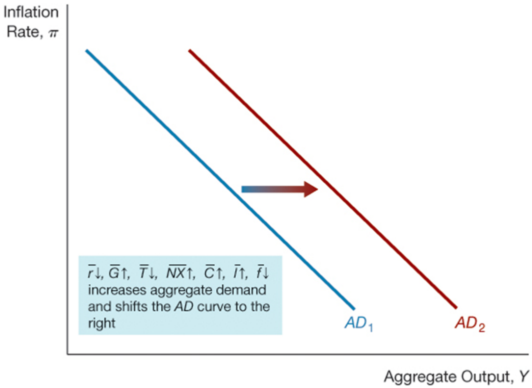
$$P \downarrow \Rightarrow \frac{M}{P} \uparrow \Rightarrow E \downarrow \Rightarrow NX \uparrow \Rightarrow Y^{ad} \uparrow$$

Shifts in the Aggregate Demand Curve



Leftward shift: lower autonomous spending or tighter monetary policy reduces output at each inflation rate.

Source: Mishkin, The Economics of Money, Banking and Financial Markets



Rightward shift: higher autonomous spending or easier monetary policy raises output at each inflation rate.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Factors That Shift the Aggregate Demand Curve

Money Supply

An increase in the money supply shifts the AD curve to the **right**.

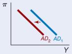
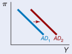
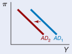
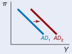
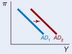
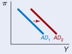
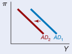
Spending Components

Higher spending in any component increases aggregate demand:

$$C,\ I,\ G,\ NX$$

Implication

More spending increases the quantity of aggregate demand at each price level.

Factor	Change	Shift in Aggregate Demand Curve
Autonomous monetary policy, $\bar{r}$	↑	
Government purchases, $\overline{G}$	↑	
Taxes, $\overline{T}$	↑	
Autonomous net exports, $\overline{NX}$	↑	
Autonomous consumption expenditure, $\overline{C}$	↑	
Autonomous investment, $\bar{I}$	↑	
Financial frictions, $\bar{f}$	↑	

Note: Only increases (↑) in the factors are shown. The effect of decreases in the factors would be the opposite of those indicated in the "Shift" column.

Factors that shift the AD curve.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Outline

Long-Run Aggregate Supply

Long-Run Aggregate Supply (LRAS)

The LRAS curve represents the economy's **potential output** (Y_P).

- Vertical at potential output (Y_P)
- Independent of the inflation rate
- Represents **full-employment output**

Determinants of LRAS

- Physical capital
- Labor at full employment
- Technology and productivity

Short-Run Aggregate Supply

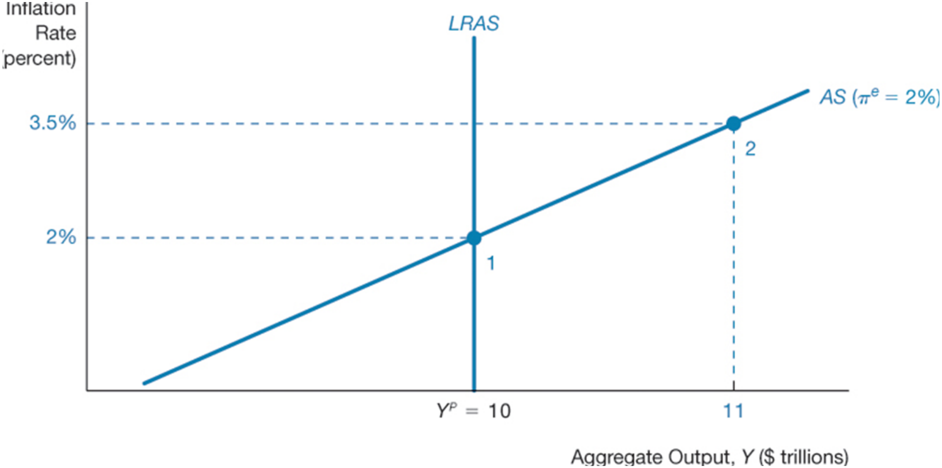
Short-Run Aggregate Supply (SRAS)

SRAS reflects the positive relationship between inflation and output in the short run due to **nominal rigidities**.

$$\pi = \pi^e + \gamma(Y - Y^P) + \rho$$

- π: actual inflation
- π^e: expected inflation
- $Y - Y^P$: output gap
- γ: sensitivity of inflation to output
- ρ: supply shock

Long- and Short-Run Aggregate Supply Curves



LRAS is vertical at potential output; SRAS slopes upward.

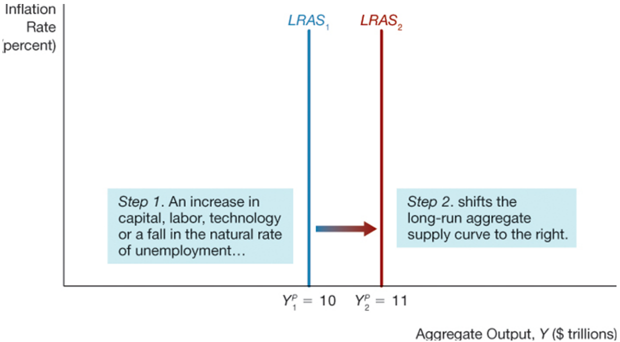
Source: Mishkin, The Economics of Money, Banking and Financial Markets

Shifts in Long-Run Aggregate Supply

Factors that Shift LRAS

- Increase in capital stock
- Increase in labor supply
- Technological progress
- Lower natural unemployment

$$Y_P \uparrow \Rightarrow LRAS \text{ shifts right}$$



Shift in the long-run aggregate supply curve.

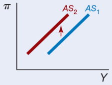
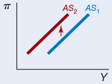
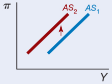
Source: Mishkin, The Economics of Money, Banking and Financial Markets

Shifts in Short-Run Aggregate Supply

SRAS Shifters

- Expected inflation
- Output gap
- Supply shocks

Each shifter enters the SRAS equation $\pi = \pi^e + \gamma(Y - Y^P) + \rho$: a rise in any of them shifts SRAS up and to the left.

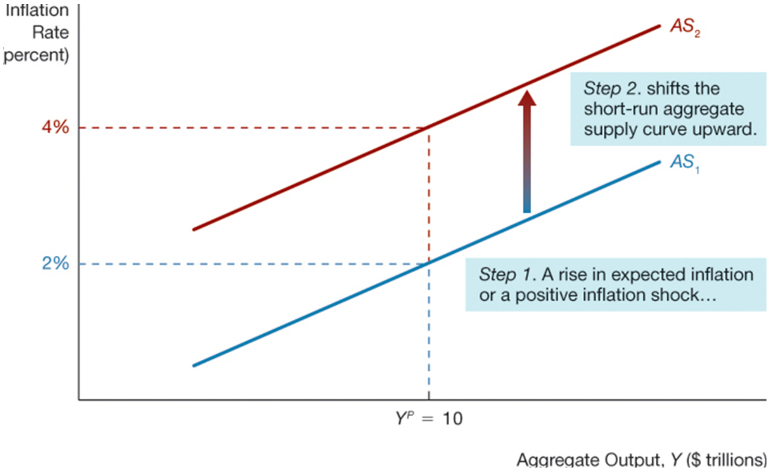
Factor	Change	Shift in Supply Curve
Expected inflation, π^e	$\uparrow$	
Inflation shock, ρ	$\uparrow$	
Persistent output gap, $(Y - Y^P)$	$\uparrow$	

Note: Only increases ($\uparrow$) in the factors are shown. The effect of decreases in the factors would be the opposite of those indicated in the "Shift" column.

Factors that shift the short-run aggregate supply curve.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

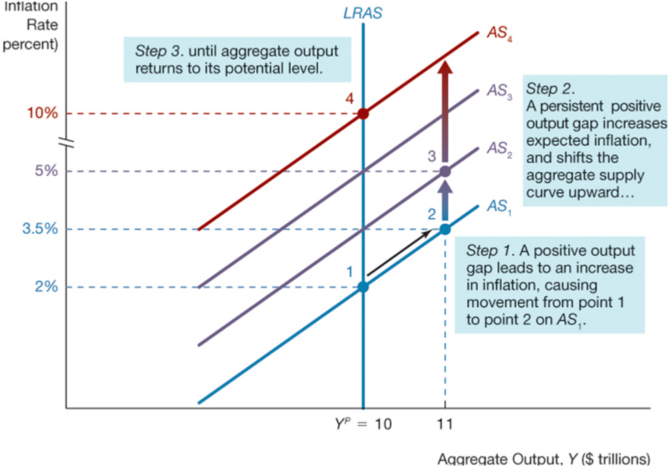
SRAS Shifts: Expected Inflation and Supply Shocks



A rise in expected inflation or a positive inflation shock shifts SRAS up.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

SRAS Shifts: A Persistent Positive Output Gap



Output above potential raises expected inflation over time, shifting SRAS up.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

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- Business Cycles and Inflation
- Aggregate Demand
- Aggregate Supply
- **Short-Run and Long-Run Equilibrium**
- Aggregate Demand Shocks
- Supply Shocks
- Conclusions from AD–AS Analysis

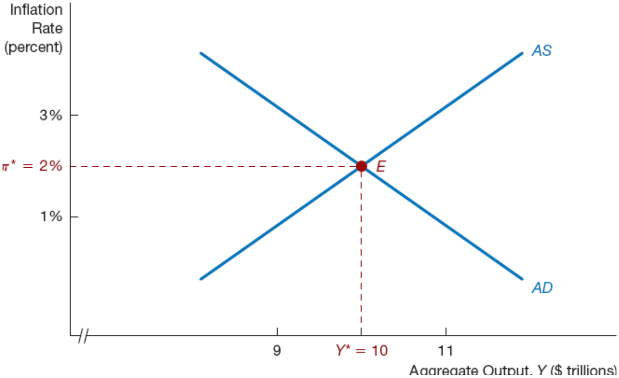
AD-AS Equilibrium

Macroeconomic Equilibrium

The economy is in equilibrium when:

$$AD = AS$$

At equilibrium, the quantity of output demanded equals the quantity supplied.



Short-run equilibrium at the intersection of AD and SRAS.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Self-Correcting Mechanism

The economy eventually returns to potential output.

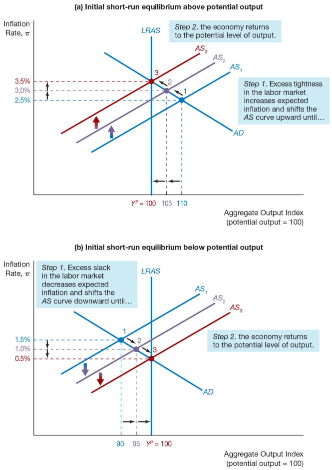
Adjustment Speed

Slow Adjustment

- Wage rigidity
- Need for policy intervention

Rapid Adjustment

- Flexible wages and prices
- Less policy intervention



Adjustment to long-run equilibrium: SRAS shifts until output returns to Y^P.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Outline

Aggregate Demand Shocks

Aggregate Demand Shock

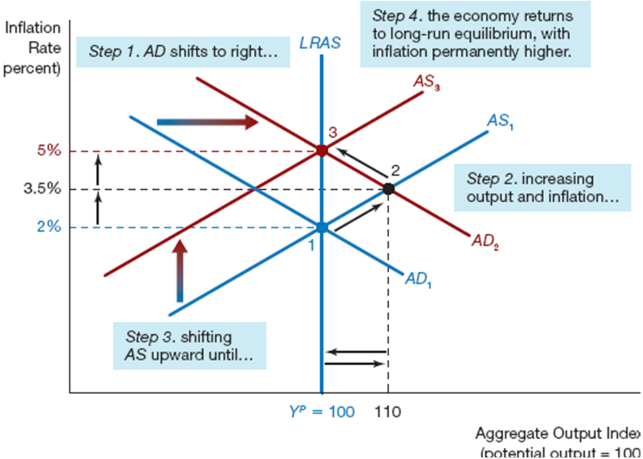
Unexpected change in spending that shifts the AD curve.

Positive AD Shocks

- Lower interest rates
- Higher government spending
- Lower taxes
- Higher consumption or investment
- Higher net exports

A positive demand shock raises output and inflation in the short run; as SRAS shifts up, output returns to potential and only inflation is permanently higher.

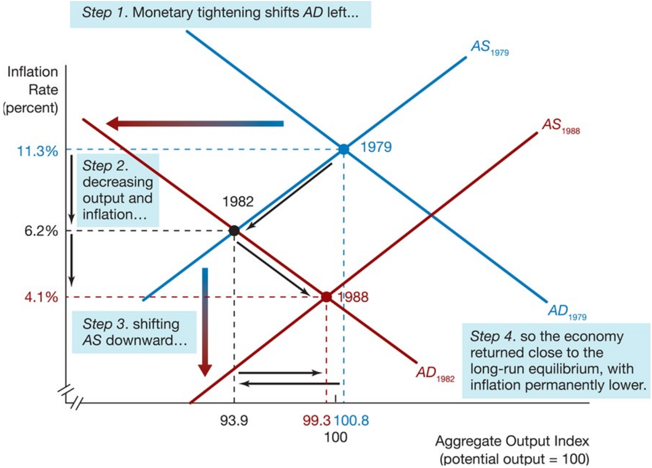
Positive Demand Shock



Response of output and inflation to a positive aggregate demand shock.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

The Volcker Disinflation



Autonomous monetary tightening shifts AD left: lower inflation, with a temporary fall in output.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Outline

Supply Shocks

Unexpected change in production costs that shifts the SRAS curve.

Negative Supply Shocks

- Oil price spikes
- Import price increases
- Wage pressures

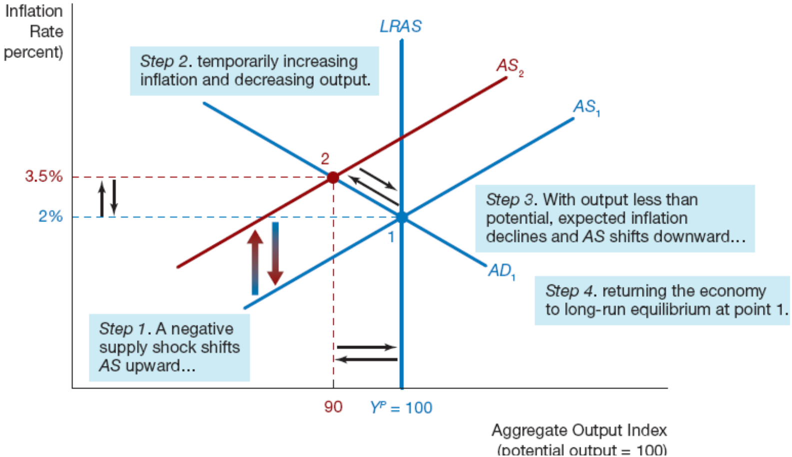
Positive Supply Shocks

- Technological progress
- Good harvests
- Falling commodity prices

Gasoline rationing in Oregon during the 1973–74 oil crisis, a classic negative supply shock.

Wikimedia Commons, David Falconer, Public domain

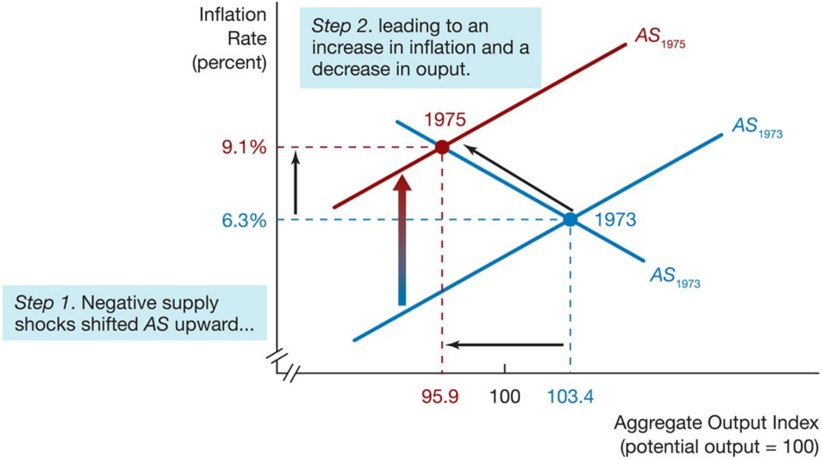
Negative Supply Shock



A temporary negative supply shock shifts SRAS up: higher inflation and lower output, then a return to potential.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Negative Supply Shocks, 1973–1975



The 1973 oil shock in the AD–AS framework: stagflation.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Permanent Supply Shocks and Real Business Cycle Theory

Permanent Negative Supply Shock

A permanent supply shock (e.g., inefficient regulations) reduces productivity and **lowers potential output**.

Impact on Aggregate Supply

- **LRAS**: shifts **left** as potential output falls
- **SRAS**: shifts **upward/left** due to higher inflation

Real Business Cycle Theory

Economists such as **Edward Prescott** argue that business cycles can be driven primarily by **permanent supply shocks**.

Edward C. Prescott (1940–2022), Nobel laureate and pioneer of real business cycle theory.

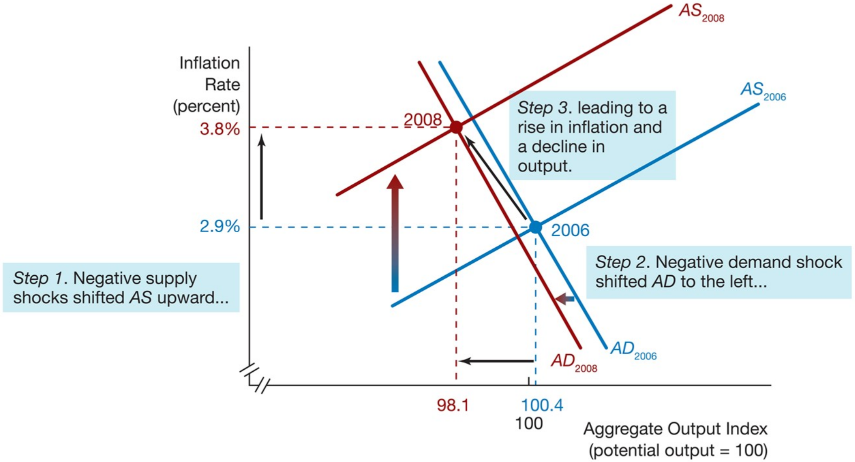
Outline

Conclusions

1. The economy has a **self-correcting mechanism** that returns output to potential and unemployment to its natural rate.
2. **Aggregate demand shocks** affect output in the **short run** but not in the long run.
3. **Temporary supply shocks** affect both output and inflation only in the short run.

Only *permanent* supply shocks change potential output and therefore have long-run effects on output; demand shocks change only inflation in the long run.

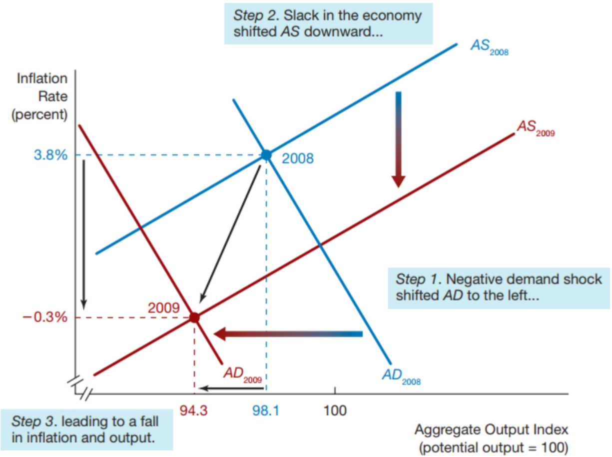
First Phase of the Great Recession



2007–08: a negative supply shock (oil) combined with a leftward AD shift.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Second Phase of the Great Recession



2008–09: the financial crisis shifts AD sharply left; output and inflation both fall.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

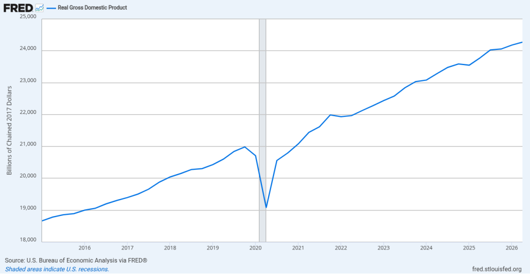
Application: AD/AS Analysis of the COVID-19 Recession (1/3)

COVID-19 Shock (2020)

Lockdowns beginning in March 2020 caused the U.S. economy to contract at the fastest rate in modern history.

Economic Impact

- Output index fell from **101.2 (2019Q4)** to **90.0 (2020Q2)**
- Unemployment rose from **3.5%** to **13%**
- Inflation declined from **2.0%** to **0.4%**



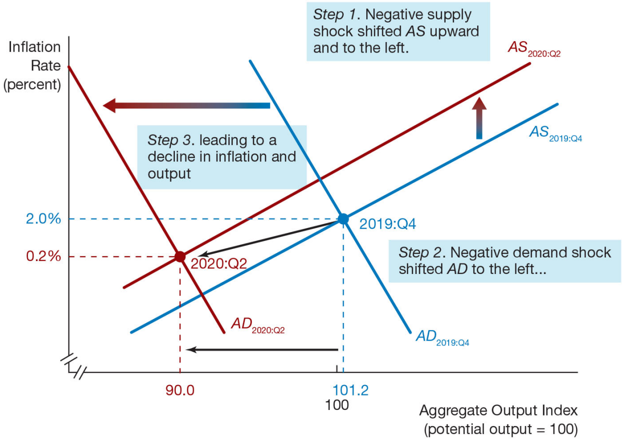
U.S. real GDP, 2015–2026: the record 2020Q2 contraction and rapid rebound.

Source: FRED, Federal Reserve Bank of St. Louis

Interpretation

The pandemic generated a major macroeconomic shock affecting output, employment, and inflation.

The COVID-19 Recession in the AD–AS Framework



A leftward SRAS shift (lockdowns) and a leftward AD shift together lowered output and inflation.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Application: AD/AS Analysis of the COVID-19 Recession (2/3)

Unusual Feature of the COVID-19 Recession

The recession was triggered by a **large supply shock** that also generated a **large demand shock**.

Supply-Side Impact

- Lockdowns in March 2020 shut down production across many sectors.
- The **SRAS** curve shifted **upward and to the left**.

Demand-Side Impact

- Lockdowns reduced household spending.
- High uncertainty discouraged consumption and investment.
- Falling stock markets further weakened spending.

Result

The **aggregate demand (AD)** curve shifted sharply **to the left**.

Application: AD/AS Analysis of the COVID-19 Recession (3/3)

Additional Event

A price war between Saudi Arabia and Russia caused oil prices to fall sharply in 2020.

Effect on Aggregate Supply

- Lower oil prices acted as a **positive supply shock**.
- This partially offset the negative supply shock from the pandemic.

Outcome

Inflation fell from about **2.0%** (2019Q4) to **0.4%** (2020Q2).

> With SRAS shifting up (lockdowns) but partly back down (cheap oil), and AD shifting far left, the net result was a large fall in output and a modest fall in inflation.

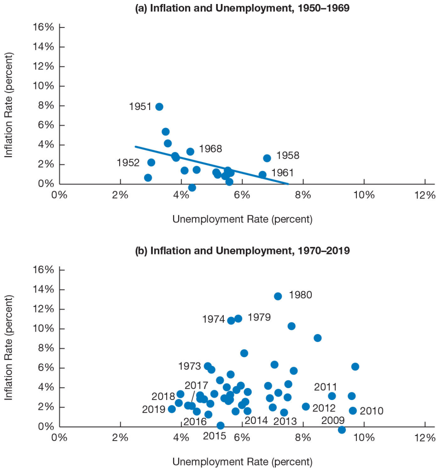
The Phillips Curve and the SRAS Curve

Phillips Curve

The Phillips curve describes the **negative relationship** between unemployment and inflation.

Intuition

- Tight labor markets (low unemployment) make hiring difficult.
- Firms raise wages to attract workers.
- Higher wages lead to higher prices and inflation.



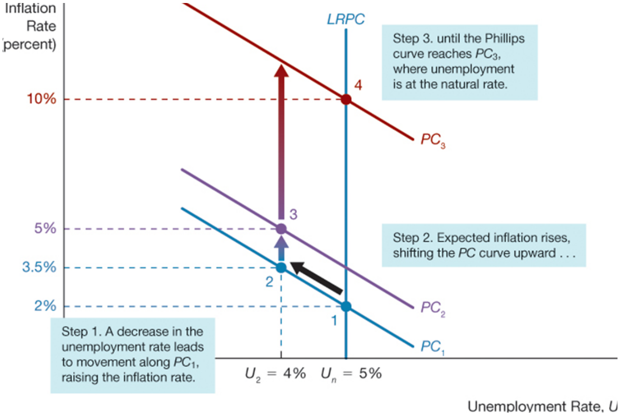
U.S. inflation and unemployment, 1950–1969 (top) and 1970–2019 (bottom): the clear trade-off of the 1950s–60s disappears later.

Source: Mishkin, The Economics of Money, Banking

The Short- and Long-Run Phillips Curve

Key Results from the Phillips Curve

1. There is **no long-run trade-off** between unemployment and inflation.
2. There is a **short-run trade-off** between unemployment and inflation.
3. The Phillips curve has two forms: short-run and long-run.



The short-run Phillips curve shifts with expected inflation; the long-run Phillips curve is vertical at the natural rate.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

The Phillips Curve After the 1960s

Expectations-Augmented Phillips Curve

The negative relationship between inflation and unemployment breaks down if unemployment stays below the natural rate for a long period.

Role of Supply Shocks

Oil price shocks in the **1970s** led economists to incorporate **inflation expectations and supply shocks** into the Phillips curve.

Adaptive Expectations

To analyze the Phillips curve, we must consider how firms and households form expectations about inflation. Under adaptive expectations, expectations are based on **past inflation** and therefore adjust only gradually over time.

From the Phillips Curve to the SRAS Curve: Okun's Law

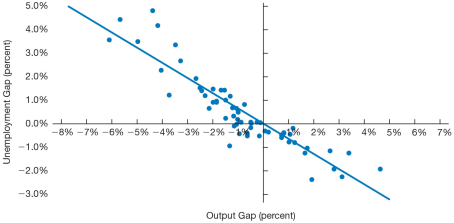
Link Between Phillips Curve and SRAS

The modern Phillips curve can be used to derive the **SRAS** curve by replacing the unemployment gap with the **output gap** $Y - Y^P$.

Okun's Law

Okun's law relates **the unemployment gap** to **the output gap**:

- If output rises above potential, unemployment falls below its natural rate.
- A 1% increase in unemployment is associated with about a **2% decrease in output** relative to potential.



Okun's law, 1960–2019: the unemployment gap against the output gap.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Chapter summary

- Business cycles are fluctuations in real GDP; the output gap and the unemployment rate measure where the economy stands, and inflation tends to rise in expansions and fall in recessions.
- The AD curve slopes downward; it shifts with the money supply and with autonomous changes in C, I, G, and NX.
- LRAS is vertical at potential output; SRAS, $\pi = \pi^e + \gamma(Y - Y^P) + \rho$, shifts with expected inflation, a persistent output gap, and supply shocks.
- A self-correcting mechanism returns output to potential: demand shocks and temporary supply shocks affect output only in the short run, while permanent supply shocks change potential output.
- The Phillips curve is a short-run trade-off only; Okun's law links the unemployment gap to the output gap and turns the Phillips curve into the SRAS curve.